IMPACT REPORTS PLATFORM — TECHNICAL GUIDANCE NOTE

Interactive Map: Provider Options & Facilities

A comparison of Google Maps API and CARTO Free API for the four map locations on the platform, prepared to support the client's decision on which mapping provider to use in production.

Prepared for: Client **Project:** Impact Reports **Date:** September 2026

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1. Where Maps Appear on the Platform

The platform currently displays an interactive map in four places:

1. **Home / Portal page** — a small preview map teasing global reach, linking to the full map page.
2. **"All Locations" page** — a full map showing one pin per country with active projects, plus a country list.
3. **Report Detail page — Countries view** — a map showing the country/countries covered by a specific report.
4. **Report Detail page — Project Sites view** — a map showing individual project locations with GPS pins for that report.

A fifth, internal-only map (the GPS picker used by staff when creating a report in the Admin Panel) is not public-facing and is excluded from the traffic estimates in this document.

2. Estimated Usage

The client has indicated expected production traffic of approximately **15,000–20,000 visitors per month**. Based on typical visitor behaviour across the map locations above (including zooming and panning), this is estimated at:

- **30,000–60,000 map views per month** (a "map view" = one map loading on a visitor's screen)
- **0.75–2.4 million map tile requests per month** (a "tile" is one small map image square; zooming/panning loads additional tiles for newly visible areas)

3. Facilities Comparison

GOOGLE MAPS API — What the client would get

- Google's own map data — roads, place names and imagery maintained directly by Google, updated frequently
- Optional satellite and hybrid (satellite + labels) map views, in addition to the standard road map
- Optional Street View integration (not part of the current design, but available if the client wants it later)
- Optional Places search / address autocomplete and turn-by-turn directions, if the platform expands to need them
- Very high brand familiarity — most visitors already know how to use a Google-style map
- Official support channel and formal Service Level Agreements available under Google's paid subscription plans
- Detailed usage dashboards and analytics via the Google Cloud Console

What it would require from the client:

- A Google Cloud billing account with a credit/debit card on file (mandatory, even though a usage tier is free)
- Ongoing monthly cost once free usage is exceeded — see Section 4
- Development work to rebuild all four map views using Google's map component (the platform's current maps are built on a different, open-source mapping library, so this is a rebuild, not a simple setting change)
- Ongoing exposure to variable billing — a sudden traffic spike (including automated bot traffic) increases the bill automatically unless a hard usage cap is configured, which would then risk the map going blank once the cap is hit

CARTO FREE API — What the client gets today

- Community-maintained OpenStreetMap map data — the same underlying map data used by Google's competitors, Apple Maps' base layer in many regions, Wikipedia, and countless government and NGO sites worldwide
- A clean, professional map style ("Voyager") already matching the platform's visual design
- All four map locations already fully built and working — country pins, project-site pins, tooltips, zoom, and pan
- Free for up to 5,000,000 map tile requests per month — comfortably above the platform's estimated usage (Section 2)
- No billing account or credit card required to stay on the free tier
- No additional development cost — this is the provider already integrated into the live platform

Current limitations:

- Standard road-map style only — no satellite imagery or Street View
- No built-in address search / autocomplete (not used anywhere on the current platform, so this has no practical effect today)
- If usage ever grows far beyond current estimates (well past 5,000,000 tile requests/month), CARTO would need to be contacted for a custom paid plan — pricing for that tier is not publicly listed and would need to be requested directly from CARTO

4. Cost Comparison at Current Traffic (15,000–20,000 visitors/month)

Provider	Free allowance	Estimated monthly cost	Billing account required?
Google Maps API	10,000 map loads/month free, then billed	**≈ $140 – $350 / month**	Yes, mandatory
CARTO Free API	5,000,000 tile requests/month free	**$0 / month**	No

Google Maps pricing: $7 per 1,000 map loads beyond the free 10,000/month allowance (Google Maps Platform pricing, 2026). Figures are estimates based on projected traffic and typical visitor map-viewing behaviour; actual usage may vary.

5. Side-by-Side Feature Summary

Facility	Google Maps API	CARTO Free API
Road map with pins & tooltips	✓ Yes	✓ Yes (already live)
Satellite / hybrid view	✓ Yes	✗ No
Street View	✓ Available	✗ No
Address search / autocomplete	✓ Available	✗ Not built in
Cost at current traffic	✗ ≈ $140–$350/month	✓ $0/month
Credit card / billing setup needed	✗ Mandatory	✓ Not required
Development effort to adopt	✗ Full rebuild of all 4 maps	✓ Already implemented
Billing risk from traffic spikes / bots	✗ Yes, uncapped by default	✓ None (free tier only)
Formal SLA / paid support	✓ Available (paid plans)	~ Only via custom/enterprise agreement

Recommendation: At the platform's current and projected traffic (15,000–20,000 visitors/month), **CARTO's free tier already covers the platform's needs at no cost**, with no change required. Google Maps would add a recurring monthly cost and a development effort to rebuild the four existing map views, without adding any facility the platform currently uses (no satellite view, Street View, or address search appears on the site today). Google Maps would only become worth the added cost and effort if the client specifically wants satellite imagery, Street View, or address-search features in the future, or if traffic grows far beyond current projections.